Foundations of Statistical Learning & Algorithms

Today's goal • 3/3 items

Module 1
Introduction to Linear Algebra and Vector Spaces

Getting Started

- Course Introduction
  Video • 1 min
- Course Overview
  Reading • 2 min
- Syllabus
  Reading • 10 min
- Meet Your Instructor
  Video • 1 min
- Meet Your Fellow Learners
  Discussion Prompt • 10 min
- Academic Integrity
  Reading • 1 min

Lesson 1: Introduction to Linear Algebra

- Introduction to Machine Learning
  Reading • 25 min
- Introduction to Linear Algebra
  Reading • 4 min
- Why Linear Algebra and Mathematics?
  Reading • 2 min
- Notation
  Reading • 5 min

Lesson 2: Systems of Linear Equations

- Foundational Concepts of Systems of Linear Equations
  Reading • 20 min
- [H5P] System of Linear Equations
  Ungraded App Item • 20 min
- Solved Example and Recommended Resources
  Reading • 20 min
- Check Your Knowledge: System of Linear Equations
  Graded Assignment • Grade: 100%

Lesson 3: Matrices

- Matrices and Matrix Operations
  Reading • 20 min
- Matrices
  Video • 8 min
- Helpful Resources and Solved Examples
  Reading • 10 min
- Check Your Knowledge: Matrices
  Graded Assignment • 16 min

Lesson 4: Vector Spaces

- Foundations of Vector Spaces: Operations and Subspaces
  Reading • 20 min
- Vector Space
  Video • 7 min
- Vector Space Properties
  Reading • 10 min
- Subspace
  Reading • 1 min
- Check Your Knowledge: Vector Spaces
  Graded Assignment • Grade: 85.71%

Lesson 5: Orthogonal Complement and Projection

- Introduction to Orthogonal Complement
  Reading • 20 min
- **Introduction to Orthogonal Projections**
  Reading • 40 min
- Importance of Projections
  Reading • 4 min
- Projection Onto One-Dimensional Subspaces (Lines)
  Reading • 15 min
- Projection Onto General Subspaces
  Reading • 10 min
- Projections Case Study
  Reading • 10 min

Module 2
Linear Transformation and Matrix Decomposition

Module 3
Probability Foundations for Statistical Learning

Module 4
Introduction to Model Evaluation and Optimization

Introduction to Orthogonal Projections

0:03 / 1:26 2x

Read Section 3.8: “Orthogonal Projections” from *Mathematics for Machine Learning* by Marc Peter Deisenroth, A. Aldo Faisal, and Cheng Soon Ong.

In this lesson, you will explore the concept of orthogonal projections, a fundamental technique in linear algebra and machine learning. Understanding orthogonal projections is crucial for various applications, including data compression, dimensionality reduction, and regression analysis.

Key Concepts:

- **Orthogonal Projections:** Learn the definition and properties of orthogonal projections in vector spaces.
- **Projection Matrices:** Understand how projection matrices are constructed and used to project vectors onto subspaces.
- **Applications:** Explore practical applications of orthogonal projections in machine learning, such as in Principal Component Analysis (PCA) and least squares regression.

As you read, consider the following questions:

1. What is an orthogonal projection, and how is it defined in the context of vector spaces?
2. How can you construct a projection matrix for a given subspace, and what are its key properties?
3. What is the significance of orthogonal projections in machine learning, and how are they used in common algorithms?
4. How do orthogonal projections help in data compression and dimensionality reduction?
5. What are some practical examples of orthogonal projections in machine learning applications, such as PCA and regression analysis?

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